Aashna Sahu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.57
Macro Vision Academy	Class XII (CBSE)	2023	92.80%
Macro Vision Academy	Class X (CBSE)	2021	95.60%

SKILLS

Languages: C++, Python, C, Go, SQL, Bash

Frameworks & Libraries: PyTorch, Keras, scikit-learn, NumPy, Pandas, Hugging Face, LangChain, FastAPI

Tools & Technologies: PostgreSQL, Prisma, REST APIs, WebSockets, Git, GitHub, Linux, WSL

Concepts: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, System Design, Machine Learning, Deep Learning, LLMs, RAG, AI Agents

WORK EXPERIENCE

Software Trainee Intern | 360 Technology

Kavalan Shield Workforce - Enterprise AI Security Governance Platform :

- Designed and implemented the network security layer of an enterprise explicit-proxy that inspects and governs employee LLM traffic to providers like OpenAI, Anthropic, and Google: built a dual-CA mutual-TLS architecture (separate device CA and interception CA) so the proxy verifies device client certificates before minting provider-domain certificates for inspection, and engineered certificate-based device identity -- in-process X.509 issuance/revocation via rcgen/OpenSSL exposed through REST APIs, with one-time key delivery and no persistence of private keys.
- Built the Rust backend (Axum, Diesel, PostgreSQL, Redis) for organization-wide security policy management, designing a Redis-only policy resolution path with zero SQL reads on the request hot path to keep enforcement low-latency, plus OIDC-based authentication (Google/Microsoft, PKCE + CSRF protection, AES-256-GCM token encryption, JWT rotation); the platform enforces allow/block/redact decisions on live traffic using a DLP engine (Microsoft Presidio) and logs enforcement activity to a TimescaleDB audit trail for compliance review.

PROJECTS

Database Engine | Go

- Built a persistent embedded database in Go with an LSM-tree storage engine: a write-ahead log with checksums and fsync-backed durability, an in-memory memtable that flushes to immutable on-disk SSTables, and background compaction that merges levels while correctly handling deletes across them. Confirmed durability with tests that kill and restart the process mid-run and check the data is still intact.
- Wrote a SQL parser and query engine from scratch supporting CREATE, SELECT, INSERT, UPDATE, and DELETE, with WHERE clauses compiled into bounded primary-key range scans instead of full table scans. Benchmarked at roughly 42K point-reads/sec and 5K range-scans/sec locally, with 20+ tests covering encoding correctness and crash recovery.

CyberTerm - P2P Encrypted Terminal Chat | Go

- Built a peer-to-peer encrypted terminal chat application in Go, implementing a custom TCP framing protocol with X25519 ECDH key exchange, HKDF-SHA256 key derivation, and AES-256-GCM authenticated encryption for confidential, tamper-evident messaging between peers.
- Designed a concurrent, event-driven terminal UI using Go channels and goroutine-based asynchronous network I/O, with all connection state changes funneled through a single-threaded update loop to keep the UI consistent under concurrent socket events.

Movie Watchlist API | Node.js, Express, PostgreSQL, Prisma

- Built a RESTful API for a movie catalog and per-user watchlists with JWT authentication, bcrypt password hashing, and Zod-validated endpoints, enforcing resource-level authorization so users can only modify records they own.
- Hardened the API with centralized error handling (mapped Prisma error codes to HTTP responses), rate limiting on auth routes, and OpenAPI/Swagger docs; covered core auth, CRUD, and ownership flows with a 22-test Jest/Supertest integration suite.

Customer Churn Prediction | Python, Pandas, Scikit-learn, XGBoost

- Built an end-to-end churn prediction pipeline on the Telco dataset (7K+ customers); handled missing/dirty data, engineered features via one-hot encoding, and benchmarked Logistic Regression, Decision Tree, Random Forest, and XGBoost, achieving 0.84 ROC-AUC.
- Diagnosed class imbalance and multicollinearity through EDA, and used correlation/coefficient analysis to identify contract type and tenure as the top churn drivers.

Adaptive Headphone Noise Canceller:

- Implemented an adaptive noise cancellation system using the *Normalised Least Mean Squares algorithm* and *Recursive Least Square Algorithm* in MATLAB.
- Achieved ~35 dB SNR Improvement, highlighting fast adaptive filtering under changing input conditions- aligns with quick decision logic in dynamic environments.

RELEVANT COURSES

Data Structures & Applications, Introduction to Artificial Intelligence, Machine Learning, Deep Learning, Probability & Random Processes, Matrix Theory, Statistical Inference,Information Theory, Convex Optimization, Error Correcting Codes